

The Quantum Approximate Optimization Algorithm

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What is QAOA?

- Quantum algorithms for **combinatorial optimization problems**
- Designed to find near-optimal solutions efficiently
- Hybrid quantum-classical algorithm

What is combinatorial optimization?

- Finding the best combination of options
- Finite, discrete set of options
- Optimize an objective function
- Often NP-hard
- MaxCut, Traveling salesman, Knapsack

$$C(z) = \sum_{\alpha=1}^m C_{\alpha}(z)$$

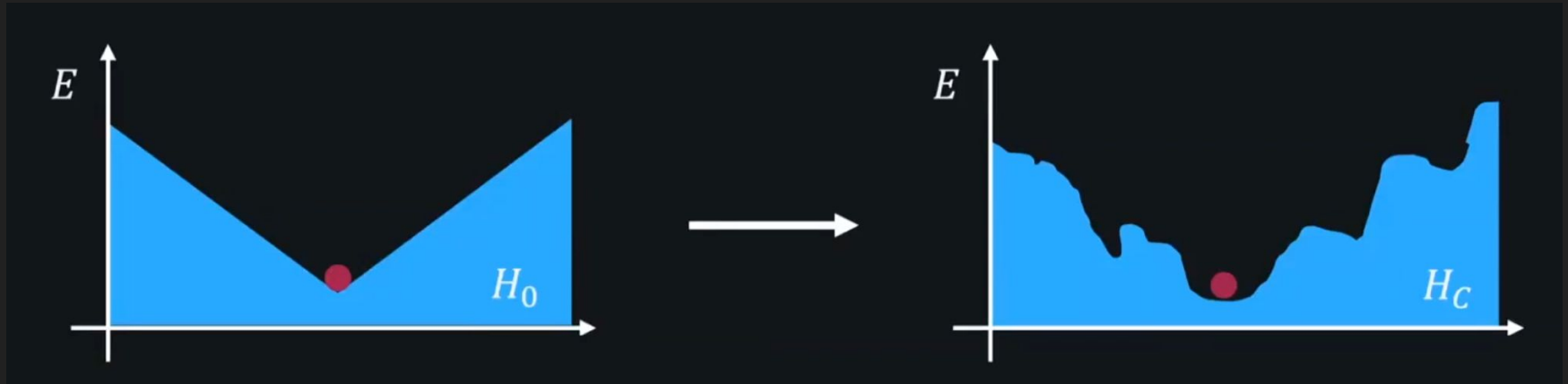
Cost Hamiltonian

- To use QAOA for combinatorial optimization problems, we need to encode it as a quantum operator
- We define that encoding as a Cost Hamiltonian

$$\hat{H}_C |\mathbf{x}\rangle = C(\mathbf{x}) |\mathbf{x}\rangle ,$$

Adiabatic computing

- Start in ground state of an 'easy' Hamiltonian
- Transform slowly into problem Hamiltonian
- Final ground state encodes the optimal solution
- System stays in lowest-energy state the whole time



QAOA trotterization

- Comes from Trotter-Suzuki formula
- Non-commuting hamiltonians are problematic to implement on a quantum computer (recall reversibility), or are simply difficult to compute
- Idea is to approximate then the non-commuting terms, as a product of a simpler terms
- Breaks the evolution into small time steps

$$\hat{U}(t) \approx \prod_{k=0}^{r-1} \exp \left[-i \hat{H}(k \Delta \tau) \Delta \tau \right] = \prod_{k=0}^{r-1} \exp \left[-i f(k \Delta \tau) \hat{H}_C \Delta \tau \right] \exp \left[-i g(k \Delta \tau) \hat{H}_M \Delta \tau \right]$$

QUBO Problems

- Quadratic Unconstrained Binary Optimization
- QUBO problems usually belong to the NP-complete class
- As name suggests, there are no constraints on the variable $\mathbf{x}$
- Find the best assignment of 0s and 1s to minimize this quadratic function

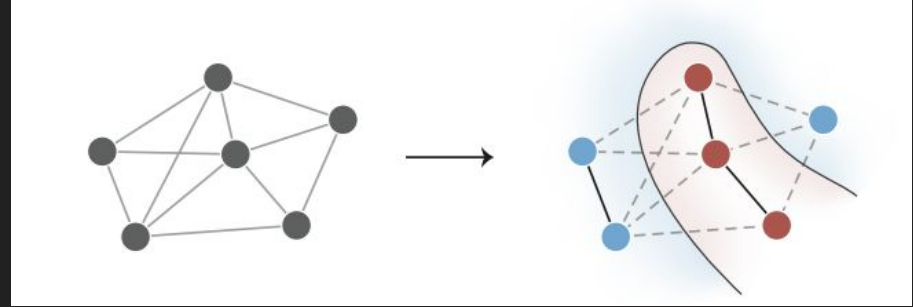
$$C(\mathbf{x}) = \mathbf{x}^T \mathbf{Q} \mathbf{x} = \sum_{i,j=1}^n Q_{ij} x_i x_j,$$

$$\mathbf{x}^* = \arg \min_{\mathbf{x} \in \{0,1\}^n} C(\mathbf{x}).$$

$$\min_{\mathbf{x} \in \{0,1\}^n} C(\mathbf{x}) = \max_{\mathbf{x} \in \{0,1\}^n} -C(\mathbf{x}) = \max_{\mathbf{x} \in \{0,1\}^n} \sum_{i,j=1}^n (-Q_{ij}) x_i x_j.$$

MaxCut problem

- Maximize weights of cut edges
- NP-hard, thus, approximation is the best approach
- We are selecting vertices to cut through, hence it is a combinatorial problem
- Can be solved with classical algorithms
- Maybe can be solved more efficiently with quantum computing?



$$C(\mathbf{x}) = \sum_{i,j=1}^{|\mathcal{V}|} w_{ij} x_i (1 - x_j),$$

Cost hamiltonian derivation for MaxCut

- Define cost Hamiltonian such that minimum energy state corresponds to the problem optimum solution
- Define mixer Hamiltonian that does not commute with HC

$$\hat{H}_C = \frac{1}{2} \sum_{(i,j) \in \mathcal{E}} w_{ij} (I - Z_i Z_j),$$
$$\hat{H}_M = \sum_{j \in \mathcal{V}} X_j,$$

QAOA for MaxCut

- 1) Init. H_C and H_M
- 2) Init. circuit in the state $|s\rangle$
- 3) Construct circuit ansatz
- 4) Define total number of layers p , and init gammas and betas
- 5) Measure final state in the computational basis
- 6) Repeat iteratively to update gammas and betas, to maximise F_p

$$|s\rangle = |+\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{\mathbf{x} \in \{0,1\}^n} |\mathbf{x}\rangle,$$

$$\hat{U}_C(\gamma) = e^{-i\gamma \hat{H}_C} = \prod_{i=1, j < i}^n R_{Z_i Z_j}(-2w_{ij}\gamma),$$

$$\hat{U}_M(\beta) = e^{-i\beta \hat{H}_M} = \prod_{i=1}^n R_{X_i}(2\beta),$$

$$|\psi_p(\gamma, \beta)\rangle = e^{-i\beta_p \hat{H}_M} e^{-i\gamma_p \hat{H}_C} \dots e^{-i\beta_1 \hat{H}_M} e^{-i\gamma_1 \hat{H}_C} |s\rangle.$$

$$F_p(\gamma, \beta) = \langle \psi_p(\gamma, \beta) | \hat{H}_C | \psi_p(\gamma, \beta) \rangle$$

$$(\gamma^*, \beta^*) = \arg \max_{\gamma, \beta} F_p(\gamma, \beta)$$

Potential Quantum Advantage

Researchers hypothesize a quantum advantage with deep circuits

QAOA has possible speedups

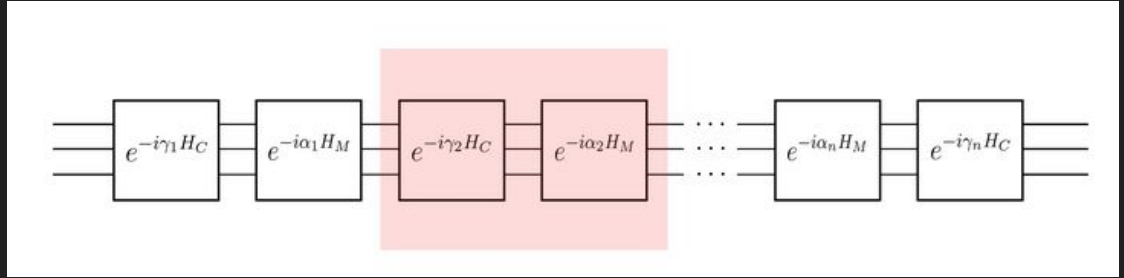
- Efficient Optimizers
- Efficient encodings and mixers
- Parallelism
- Noise Suppression

Limitations

NO GUARANTEED SPEEDUP VERSUS CLASSICAL ALGORITHMS

Implementation is difficult:

- Hardware
- Uncertainty
- Errors in measurements
- Depth
- Quantum sampling roadblock for large graphs



Time complexity depends on many variables but mostly circuit depth and problem complexity.

Approximation ratio for Max Cut: Classical (GW) - 0.878; QAOA - 0.7924
for 3-regular graphs